

Convex combinations of Channels

$$\psi = p\phi_0 + (1-p)\phi_1$$

$$\psi(x) = p\phi_0(x) + (1-p)\phi_1(x)$$

In general, $\psi = \sum_{k=0}^{m-1} p_k \phi_k$, $\psi(x) = \sum_{k=0}^{m-1} p_k \phi_k(x)$

Mixed Unitary Channels $\rightarrow \phi(\rho) = \sum_{k=0}^{m-1} p_k U_k \rho U_k^\dagger$

Pauli Channels $\rightarrow$ mixed unitary channels for which all operations are Pauli matrices